**INCLINED PLANES INTRODUCTION EASIER VERSION** **07/01/26**

A particle of mass 5kg is held at rest 10m up a rough slope angled at $20^{\circ}$ to the horizontal. It is then released and the time it takes to come to a complete stop is measured. If the particle is subject to a frictional force of 10N throughout its motion, find the time taken to come to a stop.

a) Draw a diagram showing the forces acting on the particle.
b) Find the initial acceleration of the particle.
c) Find the speed of the particle when it reaches the bottom of the slope and how long it took to slide down.
d) Draw a new force diagram for when the particle is sliding horizontally (you may omit unnecessary forces).
e) Find the time for which the particle is travelling horizontally.
f) Hence find the total time taken to come to a stop.